

Development of a System for Remote Monitoring of Patients with Chronic Diseases and an Algorithm for Assessing their Current State of Health

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Abstract—The article discusses the problems of modern monitoring of patients' health, as well as describes the principles of developing a system for remote monitoring of the health status of patients with chronic diseases and the features of forming an algorithm for evaluating the data obtained from it. The rationale for choosing a set of diagnostically significant indicators for registration is given, as well as appropriate tools are selected. A generalized structure of the remote monitoring system with a description of the modules included in it is proposed. An algorithm for analyzing medical data obtained during monitoring is considered, with the help of which the current state of the patient's health is determined. The testing of the developed monitoring system and data analysis algorithm was also carried out, according to the results of which conclusions were formulated about the effectiveness of the developed system for the tasks of remote monitoring of the health status of patients with chronic diseases.

Keywords—remote health monitoring, portable devices, noninvasive sensors, wearable sensors, control of vital signs.

I. INTRODUCTION

One of the most important problems in health care today is the monitoring of people whose health is most prone to complications and diseases. Often situations arise when patients are unable to visit their primary care physician and undergo regular diagnostics, which increases the risk of missing possible problems and future complications.

For example, with the spread of COVID-19, many pregnant women had no opportunity to be examined, not to mention already infected women, whose weakened bodies are prone to dangerous health complications for the mother and child. According to Rosstat, maternal mortality increased by 24.4% in 2020 compared to 2019 and reached 11.2% per 100,000 newborns [1].

The solution to this problem could be a system of remote monitoring of the patient's health status. It should record pre-selected diagnostically significant indicators with the ability to transmit the results of the examination to the user's or the treating physician's smartphone. It is important that the system itself can perform a preliminary analysis of the results and determine whether the values of the registered indicators are normal or whether it is necessary to pay attention and consult a doctor or whether hospitalization is even necessary.

In recent years, significant progress in the development of compact, portable biotechnical systems, research into the possibilities of wireless data transmission, increased interest in the introduction of cloud computing and data storage systems and their development have led to the fact that such systems for remote health monitoring are becoming a feasible

aspect of modern medicine [2]. The use of these systems can significantly simplify access to timely medical care, provide long-term continuous monitoring of changes in important indicators in a convenient and comfortable for patients, giving them the opportunity to manage their own health, without losing contact with the attending physician [3]. In addition, they make it possible to greatly facilitate the processes of medical care, through the organization of remote consultation and planning of treatment strategies.

The desire for a more efficient reorganization of the healthcare system has led to a huge increase in the study of remote monitoring systems and, as a consequence, the expansion of their possible application [4]. In particular, it is assumed that they can significantly improve the quality of monitoring of patients with cardiovascular [5] and respiratory [6-7] diseases, many of which are chronic.

Taking all the above into consideration, we can conclude about the relevance of designing a system that allows remote monitoring of patients with chronic forms of diseases in need of long-term health monitoring. The purpose of this work is to develop a generalized structure of such a system and an algorithm for assessing the current state of a patient on the basis of the data collected by it.

In order to achieve the goal of the study, it is necessary to solve a number of tasks:

- 1) *Determine the most important diagnostic indicators for monitoring, allowing to obtain a correct assessment of the patient's state of health;*
- 2) *To select instrumental means of acquisition and registration of the selected indicators;*
- 3) *Develop a generalized structure of remote patient health monitoring system;*
- 4) *Formulate an algorithm for assessing the current state of the patient based on the data obtained;*
- 5) *Carry out experimental testing of the developed system.*

II. MATERIALS AND METHODS

In spite of the fact that practically all remote monitoring systems will have similar structure and some component modules, each of them will differ fundamentally, depending on the final target function, a set of hardware parts and software accompanying the system. It is connected with the fact that certain algorithms, criteria of the current state evaluation and parameters of the disease exacerbation prediction are selected separately for a certain chronic

disease in order to improve the efficiency of parameters control.

In order to make the most high-quality selection of diagnostically significant parameters required for registration, prior to the introduction and use of remote monitoring system it is required to coordinate the list of parameters with the physician. If, after consultation, there are no specific instructions from the attending physician, the system will monitor the most basic and most common in all assessment techniques: heart rate, respiration, metabolism, etc.

So, for example, when monitoring pregnancy, biochemical markers of blood and urine (such as bilirubin, glucose, leukocytes, etc.) will be most important; when monitoring patients with respiratory diseases - partial pressure and external lung function indices; with cardiovascular diseases - electrocardiogram (ECG).

Depending on the set of recorded indicators determined by the doctor, the instrumental means of taking and recording a set of biomedical signals are selected. These can be wearable sensors and devices, such as ECG recording devices, portable blood pressure monitors, and smart watches. So can automatic bioanalyzers available in the patient's home, such as glucose meters, hemometers, coagulometers, inhalers, as well as indicator strips for immune-enzyme diagnostics. In addition, remote monitoring system instrumentation includes specialized cardiac monitors installed at the patient's bedside, as well as beds themselves, if equipped with, for example, elevators, and some orthopedic devices.

The main requirements for the instruments used are related to reducing their impact on the normal daily activities of the patient and, consequently, on the normal functioning of the body systems. Due to this, it is possible to minimize the errors of bio-signal registration, and therefore to increase the accuracy of the data obtained and the efficiency of remote monitoring of the patient's condition.

Fig. 1 shows a generalized structure of remote monitoring system for monitoring patients with chronic diseases, developed taking into account all of the above factors.

Several logical blocks can be conditionally distinguished in the proposed structural scheme.

The first block is responsible for remote acquisition and registration of biomedical signals and data, providing comprehensive monitoring of the patient's condition. It represents a set of medical tools (MTnN) of two subtypes a and b, providing monitoring of necessary indicators. Subtype a includes portable medical devices for home use (blood pressure monitors, glucose meters, portable cardiac monitors) and automatic biochemical analyzers (for monitoring blood and urine parameters). Subtype b includes compact wearable medical devices: sensors, trackers, and bracelets. This subtype of devices can optionally provide a connection via GSM communication channel, e.g., to a patient's relatives or directly to the attending physician (Connection Module - ConM) for cases of emergency response in case of sudden deterioration of the patient's condition.

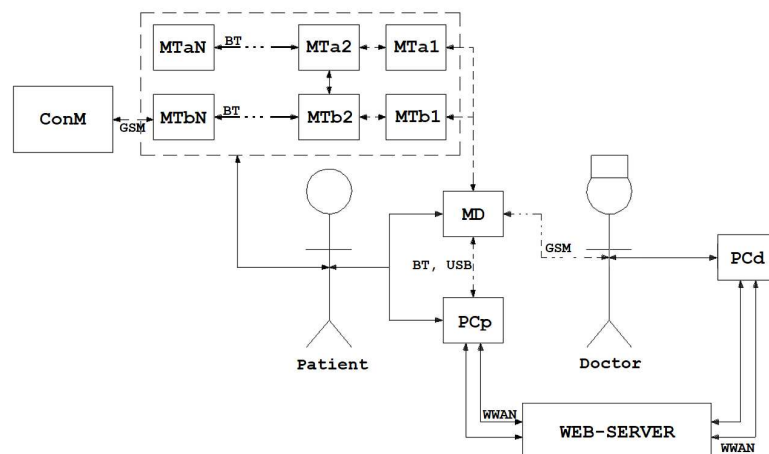


Fig. 1. Structure of proposed remote monitoring system.

All devices of this unit are connected to each other and to the next control unit via a Bluetooth data transfer channel. Recorded parameters, which in aggregate use of all individually selected medical tools allow full assessment of the patient's condition on all necessary indicators, are transmitted for further processing to a high performance smartphone or tablet of the patient (Mobile Device), for convenient storage and presentation of information having communication with the patient's personal computer (PCp).

The control unit, represented by a mobile device (MD) and a patient's computer (PCp), performs preprocessing of the taken biosignals, received from medical instruments channels, structuring of the received data, selection of the most significant indicators, which characterize the patient's health state, comparison of parameters with the values accepted as an individual norm of the patient. Analysis of the totality of the manifested factors is performed in order to

control the occurrence of disease exacerbations. We make a protocol of current values of diagnostically significant parameters and distribute them according to classes: normal, pathology and anxiety state.

Once the information has been processed, the system generates a comprehensive report on the patient's state of health, highlighting the parameters that require increased monitoring by the doctor. The finished report is transmitted via a WWAN communication channel to the medical institution's server (Web-server).

The server of the medical institution serves both as a storage unit and as a unit for additional processing of the received data, because it allows to analyze long-term changes in the functional parameters of the body and to make prognoses of possible exacerbations of the disease. In addition, information messages about critical changes in the

patient's health indicators, which can be urgently sent to the attending physician, are formed on the server of the medical institution.

For his part, the doctor can monitor information about the patient's current condition via a personal computer (PCd) at his disposal. Data from the server are sent to him in the form of specially formatted reports describing the main functional changes in the patient's body, their dynamics during long-term monitoring, as well as immediate prognoses of possible changes in the condition. In addition, the physician can be offered typical strategies for further treatment of the patient and ways to normalize his condition.

Thus, by controlling and adjusting the monitoring program in accordance with the information received, the doctor provides medical support for the patient, closing the system.

Machine learning capabilities will be used to analyze the data obtained as a result of monitoring. The main task of the analysis in this work is classification, that is, the division of the obtained data into groups. In most cases of diagnostic studies, as in this one, the processing of medical data uses two main classes: "Sick" and "Healthy". Their names may vary depending on the specific disease, its characteristics. Also, in some cases, more classes are used, for example, to clarify groups of patients. But for standard binary classification, such a tool as a decision tree - a system of algorithms for dividing information according to pre-determined features in the training sample - is perfectly suitable.

The decision tree is built from the root node (the beginning of the algorithm) through the other nodes to the so-called leaves - the ends of each of the branches. At each node, taking into account the training sample and the sample of labels (classes) for each case, by recursive partitioning of the object space so that samples with the same labels or similar target values are grouped together, and by subsequent calculation and minimization of the loss function, the rule is set, by which input data from the test sample are passed on to this or that node, thus showing all possible classification options of the analyzed information. Such algorithms are often used for predicting possible disease outcomes as well. However, special regression trees, rather than classification trees, are more often used for the prediction task.

There are several main types of decision trees depending on the learning algorithm. All other representatives are most often their refined, optimized versions. Many of them would be suitable for solving the problem of binary classification, but the most universal and at the same time simple and quite accurate is CART. It will be used for the experimental part of the study.

The algorithm for analyzing the current state of a patient on the basis of monitoring data is shown in Fig. 2.

Since the division of a set of analyzed indicators according to their belonging to a certain class is performed by comparing numerical values, it is necessary to describe the acceptable values of the analyzed attributes beforehand. In other words, the partitioning of the training sample should be carried out taking into account the normal levels of the indicators, so that it is easier for the algorithm to divide the test sample into groups. It may also be necessary to carry out additional stages of preprocessing of the input data, for example, normalization of parameter values in accordance with the standard or interpretation in a more convenient form.

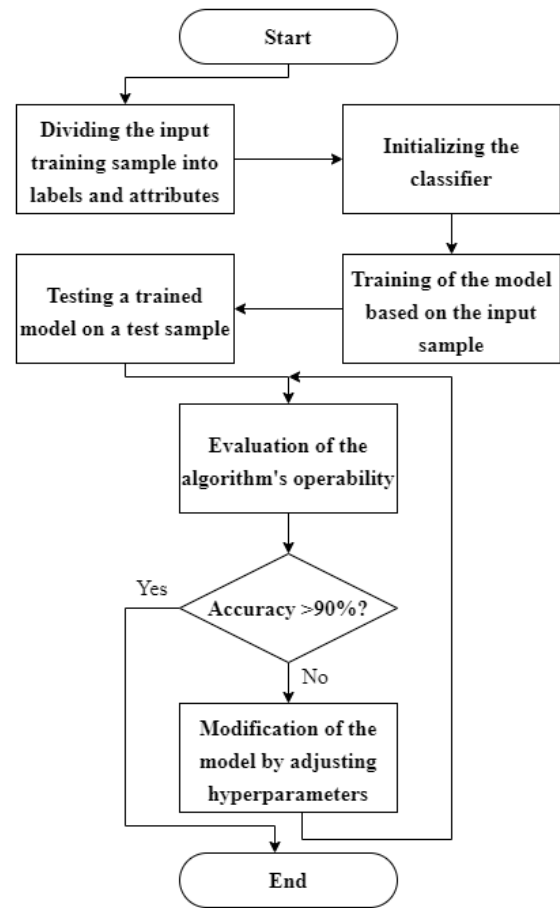


Fig. 2. Block diagram of proposed algorithm.

To start data analysis, it is necessary to divide the input training sample into labels and attributes. In other words, it is necessary to obtain two vectors, so that each set of numerical values of diagnostic parameters corresponds to a predetermined label (patient's condition).

The next stage of the algorithm is initialization of the classifier. At this stage, a decision tree model is created, ready for training. Depending on the software in which the analysis is performed, as well as on the type of learning algorithm used, this stage may be skipped or associated with the following one.

Next, it is necessary to train the built model on the input sample. Depending on the chosen training algorithm, the results of the final model, may vary, and it is at this stage you can trace the changes.

After that it is necessary to check the obtained model with a test sample. This can be pre-prepared random data or real data obtained in the course of monitoring the patient's condition. Then it is necessary to evaluate the performance of the algorithm by calculating the main criteria: accuracy, sensitivity, specificity. This can be done by different methods. In this case, it is proposed to limit ourselves to calculating the accuracy of the algorithm, which should not be less than 90%. Otherwise, the model requires fine-tuning, that is, adjusting some values of the classifier hyperparameters, which can also be set at the stage of initialization.

III. RESULTS

To test the performance of the developed system, we monitored the biochemical indices of health of several

pregnant women. Assessment of the patients' condition was proposed to be performed according to a set of 11 indicators [8] of daily urine sampling:

- 1) glucose (GLU);
- 2) bilirubin (BIL);
- 3) relative density (SG);
- 4) pH;
- 5) ketone bodies (KET);
- 6) hidden blood (BLD);
- 7) protein (PRO);
- 8) urobilinogen (URO);
- 9) nitrite (NIT);
- 10) leukocytes (LEU);
- 11) ascorbic acid (VC).

Data were obtained from daily monitoring of several pregnant women for two months. For the training sample, groups of datasets were separately selected, which were marked and divided into appropriate classes using predetermined norms of indicators. This selection was used to increase the accuracy of the algorithm, since increasing the number of samples of each group allowed more accurate classification rules to be established.

Of these data, a separate part that was not involved in the training of the model was used as a test sample. After the above analysis procedure, the following results were obtained: out of 45 data sets of the test sample, which are a set of 11 numerical values of indicators over 45 days, 12 true pathological cases were identified and 3 were identified incorrectly. The accuracy of the algorithm was approximately 93%, sensitivity 80%, and specificity 100%. That is, all normal cases were determined correctly, but pathological ones with a certain degree of error. Nevertheless, the results obtained allow us to judge a rather high accuracy of the classification algorithm, which, however, can be increased by changing the classifier hyperparameters, increasing the depth of the tree, or making changes in the training sample.

IV. CONCLUSIONS

As a result of the study the system of remote health monitoring of patients with chronic diseases and the stages of its development were described, as well as formed a set of indicators to assess the state of health, developed the structure of the system, developed an algorithm for classifying the health of patients using machine learning capabilities. Also, the developed classification algorithm was tested and conclusions about its performance were made. The obtained results will be used to finalize the obtained structure and to describe its individual blocks in more depth, as well as to direct the technical implementation of the developed remote monitoring system.

REFERENCES

- [1] Maternal death rate in Russia 2000-2021. Statista.com. <https://www.statista.com/statistics/1089661/russia-maternal-death-rate/> (accessed Nov. 5, 2022).
- [2] K. I. Mohammed *et al.*, "Real-Time Remote Health Monitoring systems: a Review on Patients Prioritisation for Multiple-Chronic Diseases, Taxonomy Analysis, Concerns and Solution Procedure," *J. Med. Syst.*, vol. 43, no. 7: 223, Jun. 2019, doi: 10.1007/s10916-019-1362-x.
- [3] A. Bertini *et al.*, "Impact of Remote Monitoring Technologies for Assisting Patients With Gestational Diabetes Mellitus: a systematic review," *Front. Bioeng. Biotechnol.*, vol. 10, Mar. 2022, doi: 10.3389/fbioe.2022.819697.
- [4] M. H. McGillion *et al.*, "Beyond Wellness Monitoring: Continuous Multiparameter Remote Automated Monitoring of Patients," *Can. J. Cardiol.*, vol. 38, no. 2, pp. 267-278, Feb. 2022, doi: 10.1016/j.cjca.2021.10.011.
- [5] J. Souza, S. Escadas and I. Baxevari, "Smart Wearable Systems for the Remote Monitoring of Selected Vascular Disorders of the Lower Extremity: A Systematic Review," *Int. J. Environ. Res. Public Health*, vol. 19, no. 22: 15231, Nov. 2022, doi: 10.3390/ijerph192215231.
- [6] G. S. Mosnaim *et al.*, "Digital Inhalers and Remote Patient Monitoring for Asthma," *J. Allergy Clin. Immunol. Pract.*, vol. 10, no. 10 pp. 2525-2533, Oct. 2022, doi:10.1016/j.jaip.2022.06.026.
- [7] I. Tomasic, N. Tomasic, R. Trobec, M. Krpan and T. Kelava, "Continuous remote monitoring of COPD patients-justification and explanation of the requirements and a survey of the available technologies," *Med. Biol. Eng. Comput.*, vol. 56, no. 4, pp. 547-569, Apr. 2018, doi:10.1007/s11517-018-1798-z.
- [8] I. P. Korneeva, K. A. Kramar, E. A. Semenova, A. M. Sergeev and Z. M. Yuldashev, "Hardware and software complex for remote monitoring and control of a pregnant woman's health state," (in Russian), *J. Inf. and Contr. Syst.*, 2021, vol. 6, no. 115, pp. 21-30, Apr. 2021, doi:10.31799/1684-8853-2021-6-21-30.